JOCE
INGENIERÍA Y MATERIALES ®

TECHNICAL CATALOG · FASTENERS

Silicon-bronze bolts

Bolts, nuts, washers and rods made of brass, silicon bronze and stainless steel, to ASTM and ANSI standards.

ASTM B99 · C-651 | ASTM B134 · B260 | ANSI B18

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01 — MATERIALS

Alloys we work with

Composition, properties and uses of each material

Brass · ASTM B134, alloy B260

This is the brass we use to make our products. It is a mix of copper, lead, iron and zinc. When zinc exceeds 18 %, the metal becomes lighter and more golden.

It is harder than pure copper. When molten it is not viscous, so it fills detailed molds perfectly. Adding 2–3 % lead and iron improves its corrosion resistance and allows tight, leak-proof connections.

Uses: fittings for pipes carrying drinking water, heating and gas distribution; condenser tubes and heat exchangers in steam power plants.



CHEMICAL COMPOSITION

Element	%
Copper	68.5 – 71.5
Lead	0.07
Iron	0.05
Zinc	Balance

Silicon bronze · ASTM alloy 651

Copper alloy with silicon, zinc, iron, manganese and lead. Silicon gives it optimal hardness, strength and conductivity.

Its mechanical properties are comparable to medium-carbon steels and its corrosion resistance is similar to copper.

Uses: electrical conductors, tanks, pressure vessels, marine construction and pressurized hydraulic lines; electrical connectors, telephone lines and oil plants. It is used most in the electrical, marine and telecommunications industries.



CHEMICAL COMPOSITION

Element	%
Copper	Balance
Silicon	0.8 – 2.0
Zinc	1.5
Iron	0.8
Manganese	0.7
Lead	0.05

02 — PRODUCT LINE

Bolts, nuts, washers and rods

Bolts, nuts, rods and washers are made in the alloys and finishes described in this catalog.

Because of the properties of the alloys we use, most of our products are made in bronze and brass. For best results, we recommend using the same alloy throughout the assembly.

Tip: one material for the whole kit

Bolt **Nut** **Washer**

AVAILABLE MATERIALS

- **Brass**
- **Bronze**
- **Stainless**



KIT Silicon-bronze bolt, washer and nut set.

03 — BOLTS

Bolts

Imperial and metric · ANSI B18.2.1 standard

A bolt is used to fasten an object. It is a cylindrical or tapered part with a helical thread and a head suited to turning it. Bolts are classified by diameter and length, head shape, thread and point. We make them in metric and imperial sizes.

Head

The part of the bolt used to turn it, by hand or with a tool (screwdriver, open-end, socket, Allen or adjustable wrench). Hex is the most requested, but we also make other shapes and drive types:

SHAPES

- Hex
- Round
- Cheese

DRIVE

- Hex
- Square
- Slotted
- Phillips
- Allen



AVAILABLE SIZES

Diameter (in)	Threads per in	Standard	Available lengths (in)
1/4	20	B18.2.1	1/2 · 3/4 · 1 · 1 1/4 · 1 1/2 · 2 · 2 1/2 · 3 · 3 1/2 · 4
5/16	18	B18.2.1	1/2 · 3/4 · 1 · 1 1/4 · 1 1/2 · 2 · 2 1/2 · 3 · 3 1/2 · 4
3/8	16	B18.2.1	1/2 · 3/4 · 1 · 1 1/4 · 1 1/2 · 2 · 2 1/2 · 3 · 3 1/2 · 4
1/2	13	B18.2.1	1/2 · 3/4 · 1 · 1 1/4 · 1 1/2 · 2 · 2 1/2 · 3 · 3 1/2 · 4
5/8	11	B18.2.1	1/2 · 3/4 · 1 · 1 1/4 · 1 1/2 · 2 · 2 1/2 · 3 · 3 1/2 · 4

AVAILABLE IN

- Brass
- Bronze
- Stainless

04 — NUTS

Nuts

ANSI B18.2.2 standard

Parts of various outer shapes with a threaded hole in the center, into which a bolt of the same diameter and thread pitch is screwed.

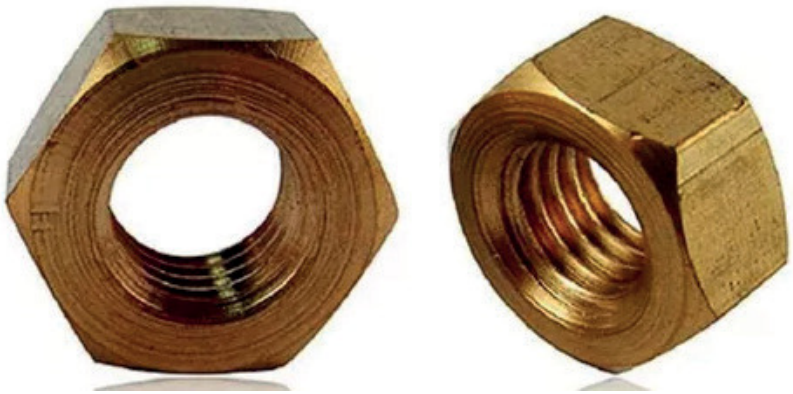
How a nut is identified

- **Number of sides:** most have 6 (hex) or 4 (square). From these basic models, many variations are possible.
- **Thickness** of the nut.
- **Bolt diameter** that fits it (the thread size, not the hole size).

TYPES

Hex · Square · Castle · Wing · Acorn · Standard





AVAILABLE SIZES · ANSI B18.2.2

Type	#5-40	#8-32	#10-24	#10-32	1/4	5/16	3/8	1/2	5/8	3/4	7/8	1
Hex	•	•	•	•	•	•	•	•	•	•	•	•
Square	•	•	•	•	•	•	•	•	•	•	•	•

AVAILABLE IN

Brass · Bronze · Stainless

05 — WASHERS

Washers

ANSI B18.21.1 standard

A washer is a thin disc with a hole, usually in the center. It is normally used to spread a clamping load; it also serves as a spacer, spring, preload indicator or locking device.

Together with the bolt and nut, the washer helps join two parts and gives a better grip and clamping between them.





TYPES

- Round (flat)
- Split lock
- Internal tooth
- External tooth
- Gasket

AVAILABLE SIZES · ANSI B18.21.1

Type	1/4	5/16	3/8	7/16	1/2	9/16	5/8	3/4
Internal tooth	•	•	•	•	•	•	•	•

AVAILABLE IN

- Brass
- Bronze
- Stainless

06 — DATA SHEET

Data sheet

Silicon bronze · copper silicon

STANDARD	ALLOY
ASTM B99	**C-651**
TEMPER	TENSILE
Half hard	**75 – 95 ksi**
MAIN USE	PRODUCTS
Electrical conduction	**Bolt, nut, washer, rod**

ksi: thousands of pounds per square inch (75–95 ksi ≈ 517–655 MPa).

CHEMICALCOMPOSITION

Element	Composition %
Copper	Balance
Silicon	0.8 – 2.0
Zinc	1.5
Iron	0.8
Manganese	0.7
Lead	0.05







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Contact us

Technical advice, quotes and custom fabrication for your next project.

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SCAN AND MESSAGE US

We answer every project.

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